Evaluating the density network

Plan: the same tests, the same measures, the same table as the smooth model

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Abstract

The fine-tuned density networks (`runs/FTD_SiLU_*_pearson`, started from `runs/DENS_L1_SiLU_thr`) look good in the fine-tune notebook's own measure. That measure is Neyman's χ^2 with an estimated number of events, which is not what the paper uses. We propose to evaluate them exactly as the smooth model was evaluated: the likelihood-ratio statistic with the exact Poisson counts, calibrated pulls with a model-error term determined on data not used in the fit, and a direct comparison of the densities. The work goes into one new notebook, `code/heft_density_eval.ipynb`, and a short results note. Decisions for you are in section 5.

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1 Why a new evaluation

Two measures of agreement with a count

Let n be the POWHEG count in a bin and μ the model's prediction, in events.

- **Neyman's χ^2**, $(n-\mu)^2/n$, uses the *observed* count as the variance. The fine-tune notebook uses this, with n estimated through N_{eff}. It favours predictions that are biased low, because a downward fluctuation gets a smaller variance and so a larger weight.
- **The likelihood-ratio statistic**, $-2\ln\lambda = 2[\mu - n + n\ln(n/\mu)]$, the logarithm of $\text{Poisson}(n;\mu)/\text{Poisson}(n;n)$, is the exact measure for counts. For a correct model

it follows χ^2_1 (mean 1). The paper uses this, with the exact counts recovered by `dihiggs/counts.py`.

A model fitted with its own Poisson variances (`VARIANCE_FROM='model'`), as the density networks were, is penalised by Neyman's measure for *not* being biased low. The comparison with the smooth model must therefore be redone with $-2\ln\lambda$.

The reference numbers on the 1,000 test spectra (mean $-2\ln\lambda$ per bin):

model	255–555 GeV	$>$ 555 GeV	median per spectrum
smooth model	1.566	1.100	1.078
free fit per bin (the floor for any 23-term model)	1.559	1.105	1.075
old network ensemble (20 networks)	1.561	1.121	1.085

2 The tests

All on the 1,000 test spectra, which played no part in training, for the L1 starting network and the four fine-tunes ($\delta = 1, 2, 3$ and plain χ^2).

2.1 Accuracy: $-2\ln\lambda$

Mean per bin below and above 555 GeV, and the median per spectrum, as in the table above; and per bin as a function of m_{hh} (the figure `fig_app_accuracy` with the density network added). The density network's prediction for a bin is its bin integral, `HEFTDensity.spectrum(theta)`.

2.2 Calibration: a model-error term for the network

The pulls need the total uncertainty of a prediction. For the smooth model this is the Poisson noise, the exact fit uncertainty, and the model-error term $\delta = \varepsilon(m)\sum_i |C_i A_i|$ (`docs/dihiggs_model.pdf`, section 6).

- **Fit uncertainty.** A single network has none. The ensembles showed that the network spread is a few per cent of the noise (5.6%), so we set it to zero for now and note this. A seeds ensemble (section 4) will supply it.
- **Model-error term.** We determine the network's own $\varepsilon(m)$ exactly as for the smooth model: on the 38,076 quick-test rows, never used in training, solve $\langle (t-f)^2/(s^2_{\text{noise}} + \varepsilon_k^2 S^2)\rangle = 1$ in each bin, with $S = \sum_i |C_i A_{ik}|$ from the network's own A_{ik}, then smooth $\ln\varepsilon$ by a quadratic. As a check we also use the smooth model's ε unchanged.

Measures: standard deviation of the pull overall and below and above 555 GeV; the fraction with $|z| < 2$; $\langle z^2\rangle$ by fifth of the cancellation ratio. The network's $\varepsilon(m)$ is itself a result: if the network sits at the same floor as the smooth model, its ε should be similar.

2.3 The density itself

- **Against the smooth model**, on a 0.5 GeV grid from 255 to 1455 GeV, for all 1,000 test points: the relative difference of $d\sigma/dm$, at the bin centres and between them, and the fraction of points within the smooth model's total $\pm1\sigma$ and $\pm2\sigma$ bands.

- **Between bin centres**: the figure `fig_interpolation` redrawn with the density network beside the old ensemble and the smooth model, for the same three spectra.
- **The 23 coefficient functions** $a_i(m)$ of the network, overlaid on the smooth model's bands (`fig_coefficients`). The poorly determined a_{16}–a_{20} may differ; the well-determined ones should agree.

2.4 Below the fitted range

With the threshold factor, the density network is defined down to 250 GeV. Its integral over 250–255 GeV is a prediction for the unused POWHEG bin 240–255 GeV (the source file holds it). The smooth model predicted that bin 8% high in total over the test spectra; we compare.

3 Implementation

- **New notebook** `code/heft_density_eval.ipynb`, in the style of the others (header, banner comments, switches). It reuses `docs/paper/smooth_common.py` for the test spectra with their exact counts, the quick-test rows, the smooth model's predictions and the old ensemble. A new helper there, `load_density_network(run)`, rebuilds a `HEFTDensity` from a run's `dihiggs_config.yaml` (activation, threshold, nodes, input) and loads its parameters.
- **Switches**: `RUNS` (the run folders to evaluate), `FEATURED` (the run shown in the figures, default `FTD_SiLU_huber_d2_pearson`).
- **Outputs**: printed tables; figures in `docs/density/` (accuracy versus m_{hh}, calibration by cancellation fifth, density comparison, coefficients, interpolation).
- **Results note** `docs/density/density_results.pdf`, in the house style.
- Runs on a laptop CPU in a few minutes; no GPU needed.

4 Later, on the 5090

1. A seeds ensemble of the featured configuration (10 seeds): the network's fit uncertainty, and its spread between bin centres on real data.
2. The merged-bin test: train on 30 GeV bins, predict the 15 GeV bins (the density model only needs 30 GeV edges; the data preparation merges pairs of rows, whose counts add).

5 Decisions for you

Decisions before I proceed

1. **Evaluate all five runs**, featuring $\delta = 2$ in the figures. Agreed, or feature another?
2. **Network fit uncertainty set to zero** until a seeds ensemble exists. Agreed?
3. **The network's own** $\varepsilon(m)$ from the quick-test rows, with the smooth model's ε as a check. Agreed?
4. **The 240–255 GeV check** included. Agreed?

5. **A new notebook** `code/heft_density_eval.ipynb` and a helper in `smooth_common.py`, rather than additions to `heft_test.ipynb`. Acceptable?